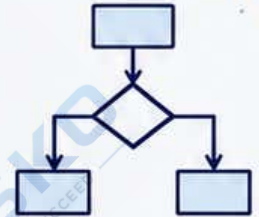


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



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Artificial Intelligence — Most Important Questions

Unit I - Introduction to AI

- **Answers in one or two lines (1 mark questions)**

1. What is Artificial Intelligence?
2. What is an AI problem?
3. What is the Turing Test?
4. What is Artificial General Intelligence (AGI)?
5. What is Super AI?
6. What is a rational agent?
7. What is an intelligent agent?
8. What is the structure of an intelligent agent?
9. What is the behavior of an intelligent agent?
10. What is a task environment?

- **Short Answer Questions (3 marks questions)**

1. What is Artificial Intelligence? Explain Artificial Intelligence and its applications.
2. What is the Turing Test? Explain the Turing Test.
3. Explain the different approaches to AI.
4. Explain Artificial General Intelligence and Super AI.
5. Explain intelligent agents, their structure and behavior.
6. Explain the task environment of an intelligent agent.

- **Long Answer Questions (6 marks questions)**

1. Explain Artificial Intelligence, its background, applications and techniques.
2. Explain different approaches to AI, including the rational agent approach.
3. Explain intelligent agents, their structure, behavior and task environment.
4. Explain the Turing Test and its significance in AI.

Unit II - Problem Solving & Searching Techniques

● Answers in one or two lines (1 mark questions)

1. What is a problem in AI?
2. What is a production system?
3. What is a control strategy?
4. What is Breadth-First Search (BFS)?
5. What is Depth-First Search (DFS)?
6. What is Hill Climbing?
7. What is a heuristic search?
8. What is Best-First Search?
9. What is A* algorithm?
10. What is a Constraint Satisfaction Problem?
11. What is Minimax?
12. What is Alpha-Beta Pruning?

● Short Answer Questions (3 marks questions)

1. What is a problem in AI? Explain problem characteristics in AI.
2. What is a production system? Explain production systems and control strategies.
3. Explain Breadth-First Search and Depth-First Search.
4. Explain Hill Climbing and its variations.
5. What is Best-First Search? Explain Best-First Search.
6. What is A* algorithm? *Explain A algorithm.**
7. Explain Constraint Satisfaction Problem (CSP).
8. Explain Minimax algorithm.
9. Explain Alpha-Beta Pruning.
10. Explain Means-End Analysis.

● Long Answer Questions (6 marks questions)

1. Explain BFS and DFS with suitable examples.
2. Explain Hill Climbing and its variations.
3. *Explain Best-First Search and A algorithm.**

4. Explain Constraint Satisfaction Problems.
5. Explain Minimax and Alpha-Beta Pruning algorithms with examples.
6. Explain production systems and control strategies in AI.

Unit III - Expert Systems

- **Answers in one or two lines (1 mark questions)**

1. What is an Expert System?
2. What is the structure of an Expert System?
3. What is a Knowledge Base?
4. What is an Inference Engine?
5. What is Knowledge Acquisition?
6. What is a Human Element in an Expert System?
7. What is an expert system shell?
8. What is machine learning?
9. What is the role of intelligent agents in expert systems?

- **Short Answer Questions (3 marks questions)**

1. 🔥 Explain the basic concepts of Expert Systems.
2. 🔥 Explain the structure of an Expert System.
3. 🔥 Explain how an Expert System works.
4. Explain the role of the human element in an Expert System.
5. 🔥 Explain expert system shells.
6. Explain problem areas addressed by Expert Systems.
7. Explain types of Expert Systems.
8. 🔥 Explain Knowledge Engineering and scope of knowledge.
9. 🔥 Explain knowledge acquisition and its methods.
10. Explain the relationship between Expert Systems, Machine Learning and Intelligent Agents..

- **Long Answer Questions (6 marks questions)**

1. 🔥 Explain Expert System, its structure and working with a suitable example.
2. 🔥 Explain types of Expert Systems and their applications.
3. 🔥 Explain Knowledge Engineering, knowledge acquisition and its methods.
4. Explain Expert Systems and their interaction with the Internet/Web.
5. Explain the role of machine learning and intelligent agents in Expert Systems.

Unit IV - Understanding Natural Languages

- **Answers in one or two lines (1 mark questions)**

1. What is Natural Language Processing?
2. What is a formal language?
3. What is a natural language?
4. What is grammar?
5. What is parsing?
6. What is semantics?
7. What is a Context-Free Grammar (CFG)?
8. What is Transformational Grammar?
9. What is a Recursive Transition Network?
10. What is the Chomsky hierarchy?

- **Short Answer Questions (3 marks questions)**

1. 🔥 Explain the components and steps of communication in NLP.
2. 🔥 Differentiate between formal and natural languages.
3. 🔥 Explain grammar, parsing and semantics.
4. 🔥 Explain the Chomsky hierarchy of grammars.
5. Explain parsing techniques.
6. 🔥 Explain Context-Free Grammar (CFG).
7. Explain Transformational Grammar.
8. Explain Recursive Transition Networks.

- **Long Answer Questions (6 marks questions)**

1. 🔥🔥 Explain Natural Language Processing and the components/steps of communication.
2. 🔥 Explain formal and natural languages with respect to grammar, parsing and semantics.
3. 🔥🔥 Explain the Chomsky hierarchy of grammars in detail.
4. 🔥 Explain Context-Free Grammar and parsing techniques.
5. Explain Transformational Grammar and Recursive Transition Networks.